MitoChime: A Machine Learning Pipeline for Detecting PCR-Induced Chimeras in Mitochondrial Illumina Reads

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Abstract. Polymerase chain reaction (PCR) amplification during Illumina library preparation can introduce chimeric reads that degrade mitochondrial genome assembly by creating spurious junctions and fragmented assembly graphs. Existing chimera detection tools are largely optimized for amplicon-based microbial workflows and often rely on reference databases or abundance assumptions that do not fit single-species organellar sequencing. We present a pipeline for detecting PCR-induced chimeric reads in *Sardinella lemuru* mitochondrial paired-end data using alignment-derived features and a sequence-based convolutional model. Using simulated clean and chimeric reads, alignment-structure and clipping metrics are extracted alongside k-mer discontinuity and microhomology descriptors as features. On a held-out test set, a tuned Gradient Boosting classifier achieves F1=0.777 and ROC–AUC=0.846 for chimeric-read detection, while a 1D Convolutional Neural Network model achieves F1=0.914 and ROC–AUC=0.961 with higher recall. Assembly evaluation using SPAdes-generated assembly graphs summarized with Bandage statistics shows that filtering reduced assembly graph complexity at high chimera rates, with performance depending on read depth. These results support MitoChime as a practical pre-assembly quality-control step for mitochondrial genome assembly pipelines.

Keywords: Chimera detection · Mitochondrial genome assembly · Illumina reads · Machine learning · Deep learning

1 Introduction

Polymerase chain reaction (PCR) amplification during next-generation sequencing (NGS) library preparation can generate chimeric reads, which are artificial recombinant sequences formed when partially copied DNA fragments attach to different templates and are extended in later cycles [10,13]. As illustrated in Fig. 1, incomplete products from two templates can combine and then be amplified into one artificial recombinant sequence. Despite the high throughput and accuracy of Illumina sequencing, PCR-induced chimeras remain a persistent artifact that can affect downstream analyses and compromise genome assembly.

Mitochondrial genomes are particularly susceptible to this issue because they are small, circular, and often contain repetitive regions [2,4]. Under some conditions, junction-supporting artifacts may introduce misleading connections in

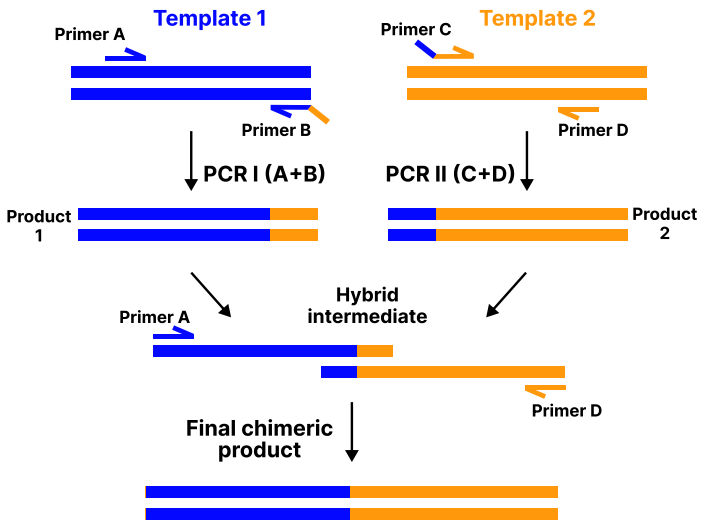

Fig. 1: Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching, adapted from [16].

assembly graphs, produce branching structures or fragmentation, or in some cases contribute to assembly failure [5,8,9]. Because many organelle assembly tools such as SPAdes [1], GetOrganelle [9], and MITObim [8], and assume that input reads are largely free of such artifacts, chimera-related assembly failure modes remain insufficiently addressed.

Many existing chimera detection approaches were originally developed for amplicon-based microbial community sequencing and often use differences in sequence frequency to help identify chimeras [7,15]. Although effective in many microbial studies, these approaches are not well suited to single-species organellar sequencing, highlighting the need for a more specialized method. A pipeline that detects chimeric reads from alignment and sequence irregularities can therefore serve as a useful pre-assembly filtering step.

Sardinella lemuru was selected as the target organism because it is an economically important sardine species in the Philippines and a relevant subject for mitochondrial genome analysis [11,17]. Its selection was also supported by the availability of mitochondrial assemblies and raw sequencing datasets in public repositories such as the National Center for Biotechnology Information (NCBI), which facilitates reference selection and benchmarking. This study introduces MitoChime, a pipeline for detecting PCR-induced chimeric reads in *Sardinella lemuru* mitochondrial Illumina paired-end data. The pipeline extracts interpretable features from standard BAM and FASTQ artifacts, benchmarks supervised machine learning models together with a sequence-based convolutional

neural network (CNN), and evaluates the effects of filtering on downstream assembly graph complexity.

2 Related Work

Chimera detection in NGS has been widely studied in amplicon sequencing, where tools such as UCHIME and UCHIME2 detect sequences that appear to be recombined from two parent sequences [7,6]. DADA2 detects chimeras in amplicon data at the level of inferred sequence variants rather than from irregularities in individual reads [3]. Although effective in many microbial studies, these approaches are not designed for read-level detection of PCR-induced chimeras in single-species organellar sequencing data.

Ensemble approaches such as CATCh use the scores and binary decisions produced by multiple chimera detection tools as input to a supervised learning model, improving detection in 16S rRNA amplicon sequencing but still relying on the sequence and alignment signals examined by those tools [12]. Pipelines such as ChimPipe detect biological chimeras in RNA-seq using discordant paired-end and split-read evidence, but are intended for fusion detection and usually require reference genome and annotation data [14].

In contrast, mitochondrial assembly pipelines generally do not provide explicit read-level chimera filtering tailored to PCR artifacts. This gap motivates a feature-driven approach that targets alignment fragmentation and sequence discontinuity in mitochondrial paired-end reads and evaluates success in terms of downstream assembly graph outcomes.

3 Methods

3.1 Overview

Figure 2 summarizes the MitoChime workflow: paired-end read simulation, alignment to a mitochondrial reference, (i) tabular feature extraction for classical machine learning and (ii) sequence-only dataset preparation for deep learning, followed by model inference, pair-safe read filtering, and downstream assembly evaluation.

3.2 Data generation and alignment

Paired-end Illumina reads were simulated from the *Sardinella lemuru* mitochondrial reference (NC_039553.1). A total of 40,000 paired-end reads were generated, comprising 20,000 chimeric and 20,000 clean read pairs. Clean read pairs were generated with `wgsim` (v1.20) at 150 bp using an error rate of 0.05. Chimeric templates were constructed by concatenating two non-adjacent reference segments, optionally retaining short overlaps to approximate PCR template switching, and were simulated with the same read-length configuration but with

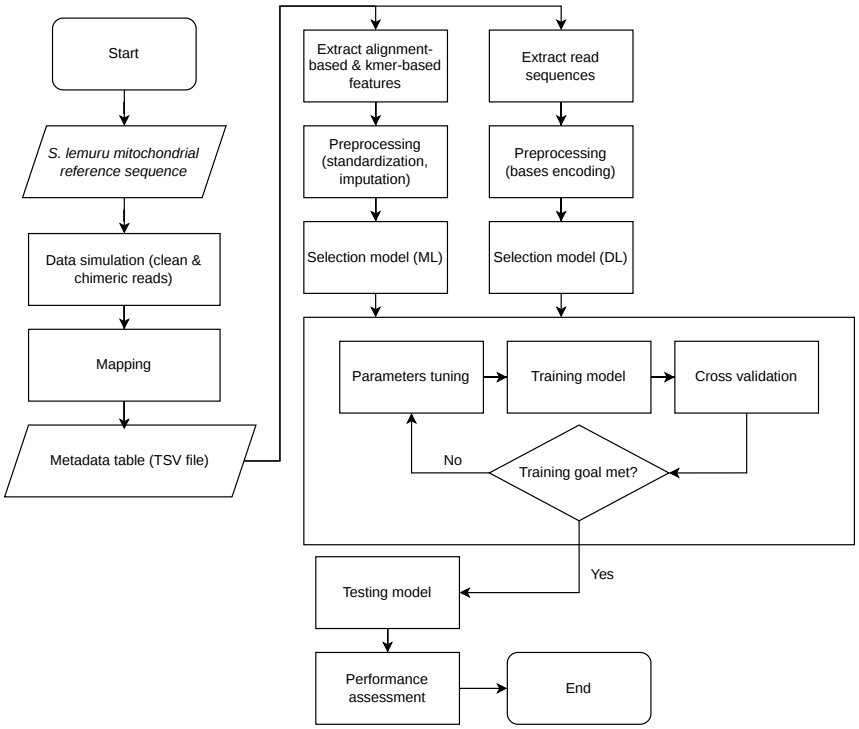

Fig. 2: MitoChime workflow from simulated reads to filtering and assembly evaluation.

the error rate set to 0 to preserve junction signals. Reads were aligned to the reference using `minimap2` (v2.30-r1287) in short-read mode (`-ax sr`) and processed with `samtools` (v1.20) to produce sorted and indexed BAM files. An illustrative example of one read from the dataset is shown in Figure 3.

Example data point from the dataset

```
read_id: NC_039553.1_10001_10492_5:0:0_8:0:0_b77/1
label: 0
seq: GTAAGGCATAGTAAGGCAATGGTGAGGCTGTGTGTGATTGCAGCAGTTCAAATTCACT...
```

Fig. 3: Illustrative example of one record from the simulated dataset. The sequence is truncated for readability.

3.3 Feature and sequence extraction

Two complementary input representations were prepared from the same labeled read pairs to support both classical machine learning and deep learning.

For classical machine learning, a tabular feature matrix was constructed where for each primary-mapped read, a fixed-length feature vector was computed using `pysam` and NumPy. Feature families included: (i) supplementary-alignment and alignment-structure summaries derived from SA tags (e.g., presence/count of supplementary alignments, segment distances, strand consistency, and MAPQ/NM aggregates); (ii) CIGAR-derived clipping statistics (e.g., left/right soft clipping, total clipped bases, and an inferred breakpoint position); (iii) k-mer composition discontinuity between read segments ($k = 6$), quantified via cosine distance and Jensen–Shannon divergence between left/right k-mer profiles; and (iv) microhomology descriptors around the inferred breakpoint (longest exact suffix–prefix overlap within a ± 40 bp window and its GC fraction), together with basic alignment-quality indicators.

In parallel, for deep learning, the raw nucleotide sequence (length $L = 150$) was retained as a separate input for the CNN. Sequences were one-hot encoded at training and inference time, with four base channels (A, C, G, T). Ambiguous bases were represented as all zeros.

3.4 Modeling and evaluation protocol

The labeled dataset was partitioned into training and test splits (80/20) using stratification to preserve the balanced class ratio. Model selection, tuning, and five-fold cross validation were performed on the training split only. Final performance was reported on the held-out test split using accuracy, precision, recall, F1, and ROC–AUC (precision/recall/F1 for the chimeric class).

Classical machine learning (feature-based). Feature-based models were implemented in Python using `scikit-learn`, `xgboost`, `lightgbm`, and `catboost`. An initial benchmark evaluated a panel of classifiers under five-fold stratified cross-validation on the training split, including a dummy baseline, L2-regularized logistic regression, a calibrated linear SVM, k-nearest neighbours, Gaussian Naïve Bayes, tree ensembles (Random Forest, ExtraTrees, and bagging with decision trees), and gradient boosting methods (Gradient Boosting, XGBoost, LightGBM, and CatBoost). Based on cross-validation ranking, a subset of competitive model families was carried forward for hyperparameter optimization. Hyperparameters were tuned using randomized search with five-fold stratified cross-validation (`RandomizedSearchCV`), optimizing chimeric-class F1. Among the feature-based models, a tuned `GradientBoostingClassifier` (`sklearn.ensemble`) achieved the best validation ranking and was used as the primary classical model. The hyperparameter search included the number of estimators, `learning_rate` (selected value 0.2), and tree-structure constraints; the selected model was refit on the full training split and evaluated once on the held-out test split.

Deep learning (sequence-based). A one-dimensional convolutional neural network (CNN1D) was trained using sequence-only input of length $L = 150$, represented by one-hot encoding. The architecture comprised three convolutional blocks with 64, 128, and 256 filters (kernel sizes 7, 5, and 3; ReLU activations; max-pooling after the first two blocks), followed by adaptive max pooling and a two-layer MLP head ($256 \rightarrow 128 \rightarrow 2$) with dropout (0.2). Training used mini-batches of 128 for up to 30 epochs with learning rate 0.001 and weight decay 10^{-4}. The selected checkpoint was the model achieving the best held-out F1 during training.

3.5 Filtering strategy

Filtering was performed using model scores for the chimeric class and a user-specified decision threshold $\tau \in [0, 1]$. For each read pair, the predicted chimeric probability p was computed and pairs were retained if $p < \tau$ (i.e., higher τ yields stricter chimera removal). Pair-safe filtering was enforced by normalizing identifiers to a base read-pair ID: leading `@` characters and trailing mate suffixes (`/1`, `/2`) were removed, and unique base IDs were expanded to mate-specific identifiers by writing both `base/1` and `base/2`. Filtered FASTQ outputs were then produced by retaining only R1 records whose base IDs passed the filter and writing only those R2 records with matching base IDs, ensuring consistent retention/removal of both mates.

3.6 Assembly and downstream evaluation

Assemblies were generated from unfiltered and filtered read sets using SPAdes. Assembly graphs were summarized using Bandage graph statistics. Graph complexity was quantified using graph-level metrics including node and edge counts, together with total assembled length, across simulated read depths and chimera rates.

3.7 Chimeric Detection Pipeline

A command-line interface automates preprocessing, model inference, pair-safe filtering, and optional assembly evaluation (Fig. 4) to support reproducible use in mitochondrial assembly workflows.

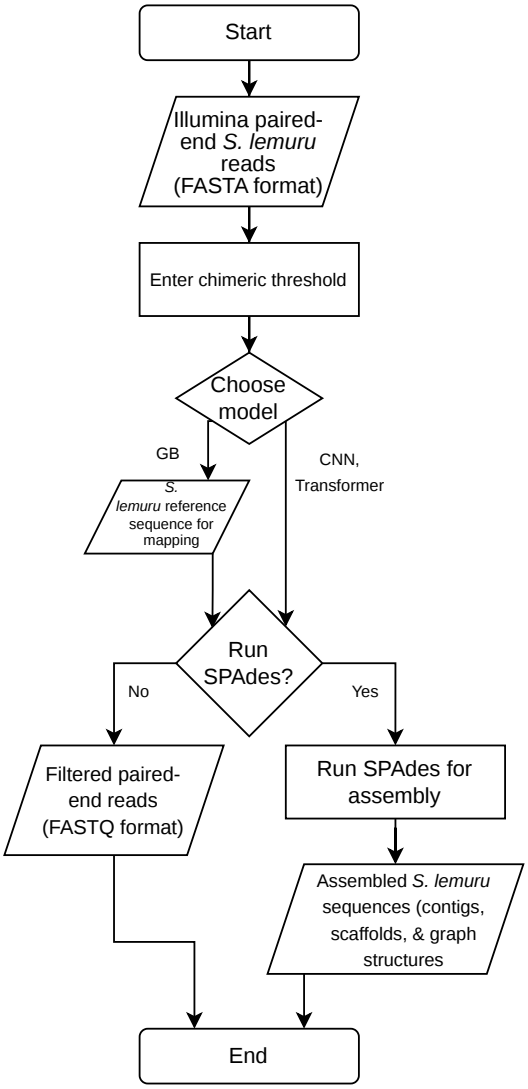

Fig. 4: MitoChime pipeline for model selection, filtering, and optional assembly evaluation.

4 Results

4.1 Held-out test performance

Table 1 summarizes held-out test performance for the tuned Gradient Boosting (feature-based) classifier and the CNN1D (sequence-based) model. Precision/recall/F1 are reported for the chimeric class. The tuned Gradient Boosting model exhibits high precision (0.937) but lower recall (0.663), indicating a conservative decision strategy that reduced false positives at the expense of missed chimeric reads. In contrast, the CNN1D substantially increases recall (0.959) and overall discrimination (ROC–AUC 0.961), at the cost of reduced precision (0.872), resulting in more clean reads flagged as chimeric.

Table 1: Held-out test performance of the tuned Gradient Boosting model and CNN1D. Precision/recall/F1 are reported for the chimeric (positive) class.

Model	Accuracy	Precision	Recall	F1	ROC–AUC
Gradient Boosting (tuned)	0.8091	0.9368	0.6630	0.7765	0.8459
CNN1D (final)	0.9093	0.8724	0.9588	0.9135	0.9607

4.2 Error profiles

The trade-off between the two models is visible in Fig. 5. Gradient Boosting produces relatively few false positives (179 clean reads predicted as chimeric) but many false negatives (1,348 chimeric reads predicted as clean). CNN1D reverses this pattern, with markedly fewer false negatives (165) but more false positives (561). These complementary error profiles motivate evaluating both models in terms of downstream assembly outcomes rather than classification metrics alone.

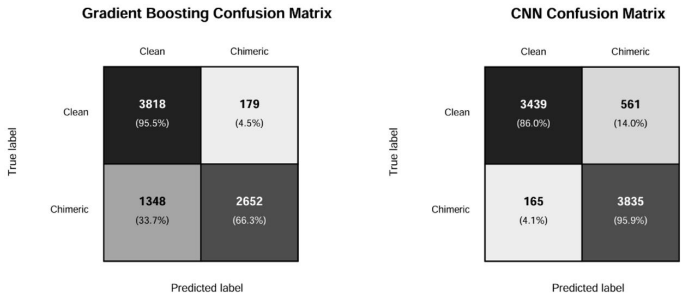

Fig. 5: Confusion matrices for the Gradient Boosting and CNN1D models on the held-out test.

4.3 External Validation Under Varying Read Depths and Chimera Rates

Table 2 reports Bandage and SPAdes graph statistics for unfiltered and filtered assemblies across read depths and chimera rates. Filtering substantially reduces graph complexity under high chimera rates. At 20k reads and 50% chimeras, the unfiltered assembly becomes extremely fragmented (966 nodes), whereas both GB and CNN1D restore near-contiguity (3 and 2 nodes, respectively). At 10k reads and 50% chimeras, CNN1D collapses the graph to a single node, while GB reduces complexity to 2 nodes, consistent with CNN1D's higher chimera recall and more aggressive removal of ambiguous read links. However, at low depth (3.2k), CNN1D occasionally increases fragmentation at higher chimera rates (15–50%), suggesting that over-filtering can remove critical read support when coverage is marginal. Overall, GB provides a conservative approach for low-depth inputs, while CNN is most beneficial at moderate-to-high coverage and high chimeric rates.

Table 2: Assembly graph statistics (Bandage and SPAdes) for unfiltered vs. filtered reads across read depths and chimera rates. Nodes summarize graph complexity (lower is simpler).

Depth	Chimera%	Condition	Nodes	Edges	Total length
3.2k	5	Unfiltered	2	0	16,714
	5	GB	1	0	16,577
	5	CNN	1	0	16,352
	10	Unfiltered	1	0	16,591
	10	GB	1	0	16,615
	10	CNN	1	0	16,584
	15	Unfiltered	1	0	16,645
	15	GB	1	0	16,607
	15	CNN	2	0	16,438
	50	Unfiltered	1	0	16,736
	50	GB	1	0	16,646
	50	CNN	2	0	16,714
10k	5	Unfiltered	1	0	16,615
	5	GB	1	0	16,615
	5	CNN	1	0	16,599
	10	Unfiltered	1	0	16,704
	10	GB	1	0	16,614
	10	CNN	1	0	16,614
	15	Unfiltered	1	0	16,735
	15	GB	1	0	16,642
	15	CNN	1	0	16,611
	50	Unfiltered	3	4	16,945
	50	GB	2	2	16,902
	50	CNN	1	0	16,651
20k	5	Unfiltered	1	0	16,669
	5	GB	2	2	16,734
	5	CNN	1	0	16,669
	10	Unfiltered	3	4	17,029
	10	GB	1	0	16,616
	10	CNN	1	0	16,770
	15	Unfiltered	3	4	16,977
	15	GB	1	0	16,891
	15	CNN	1	0	16,666
	50	Unfiltered	966	1,297	163,906
	50	GB	3	4	17,029
	50	CNN	2	2	16,808

5 Discussion

Results show that PCR-induced chimeras in short-read mitochondrial data can be detected and effectively removed using interpretable features derived from standard alignments, and that a sequence-based CNN can further improve sensitivity. The Gradient Boosting model is conservative, achieving high precision for the chimeric class but missing more chimeric reads. The CNN1D substantially improves recall and ROC-AUC, but also increases the number of clean reads incorrectly flagged as chimeric, which can reduce read coverage when the read depth is low. Downstream assembly evaluation indicates that optimal filtering depends on both coverage and chimera percentage. At moderate-to-high coverage and high chimera rates, highly sensitive chimeric filtering can simplify graphs and restore near-contiguity. At minimal depth, however, aggressive filtering may remove critical read support and introduce fragmentation. These

findings suggest that deployment should show a tunable threshold and report read-retention diagnostics to support user calibration.

6 Conclusion

MitoChime offers a practical pre-assembly chimera detection pipeline for mitochondrial Illumina paired-end reads using interpretable alignment-based and sequence-derived signals. The tuned Gradient Boosting classifier achieves robust performance with conservative filtering, while a CNN1D model provides higher sensitivity and stronger overall discrimination. External assembly evaluation further shows that filtering can substantially reduce assembly graph complexity under high chimera percentage, with coverage-dependent trade-offs at low depth. In general, the Gradient Boosting model is better suited to low-coverage datasets where preserving read support is critical, while the CNN1D is more advantageous at moderate-to-high coverage where its aggressive chimera detection can be fully utilized without risking assembly fragmentation. These trade-offs may also be adjusted through model selection and tuning depending on the characteristics of the input data and the specific goals of the user. Overall, these findings support MitoChime as a useful quality-control step for improving mitochondrial genome assembly workflows, particularly in resource-constrained settings.

Limitations include reliance on simulation and a single target species. Broader validation across taxa, library preparation conditions, and empirical datasets is required to establish generalizability. Future work includes calibration on real sequencing libraries, evaluation under varying sequencing error profiles, and extension of the pipeline to support additional model families and references

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